Developer Technology Trends: Today and Tomorrow

IBM Data Analyst Professional Certificate · Capstone Project

Mounika Singam

October 18, 2026

Outline

3	Executive Summary
4	Introduction
5	Methodology
6–7	Results: Programming Language Trends
8–9	Results: Database Trends
10	Job Postings by Location
11–14	Dashboard: Current Technology Usage, Future Technology Trends, Demographics
15	Dashboard Insights
16	Overall Findings & Implications
17	Conclusion
18	Appendix

Executive Summary

- **Who answered:** 18,845 developers from 161 countries; 91% code professionally.
- **Languages today:** JavaScript (79%), SQL (67%) and HTML/CSS (66%) lead; TypeScript and Python follow.
- **Languages next year:** Go and Rust enter the top 10. Rust draws 2.45 “want” votes per current user, the highest among languages with 1,000+ users.
- **Databases:** PostgreSQL leads today (61%) and widens its lead next year (65%). Redis climbs to #2.
- **Cloud & frameworks:** AWS dominates (58%). React and Node.js lead web frameworks; Svelte and FastAPI are rising.
- **Demographics:** 69% are aged 25–44 and 72% hold a Bachelor’s or Master’s degree. The US, Germany and India are the largest groups.

18,845

survey respondents analysed

79%

have used JavaScript

2.45×

Rust: future demand vs. current use

Introduction

Technology changes fast. This report uses survey data to show which skills developers rely on today and which they plan to adopt next.



Purpose

Identify the most-used and most-wanted programming languages, databases, cloud platforms and web frameworks, and profile the respondents behind them.



Target audience

Hiring managers, learning & development teams, technology leaders, and aspiring developers and data analysts planning their careers.



Value

Evidence-based guidance on where to invest in hiring, training and tooling, and on which skills will stay in demand next year.

Methodology

1 Data sources

- Stack Overflow Developer Survey 2024: online questionnaire, subset of 18,845 responses × 114 columns (IBM Skills Network)
- Job postings: collected via API / web scraping (job-postings.xlsx)

2 Data wrangling

- Checked duplicates: 0 duplicate rows or IDs
- Checked missing values: none in tech or demographic fields
- Split ";" multi-select answers into one row per technology

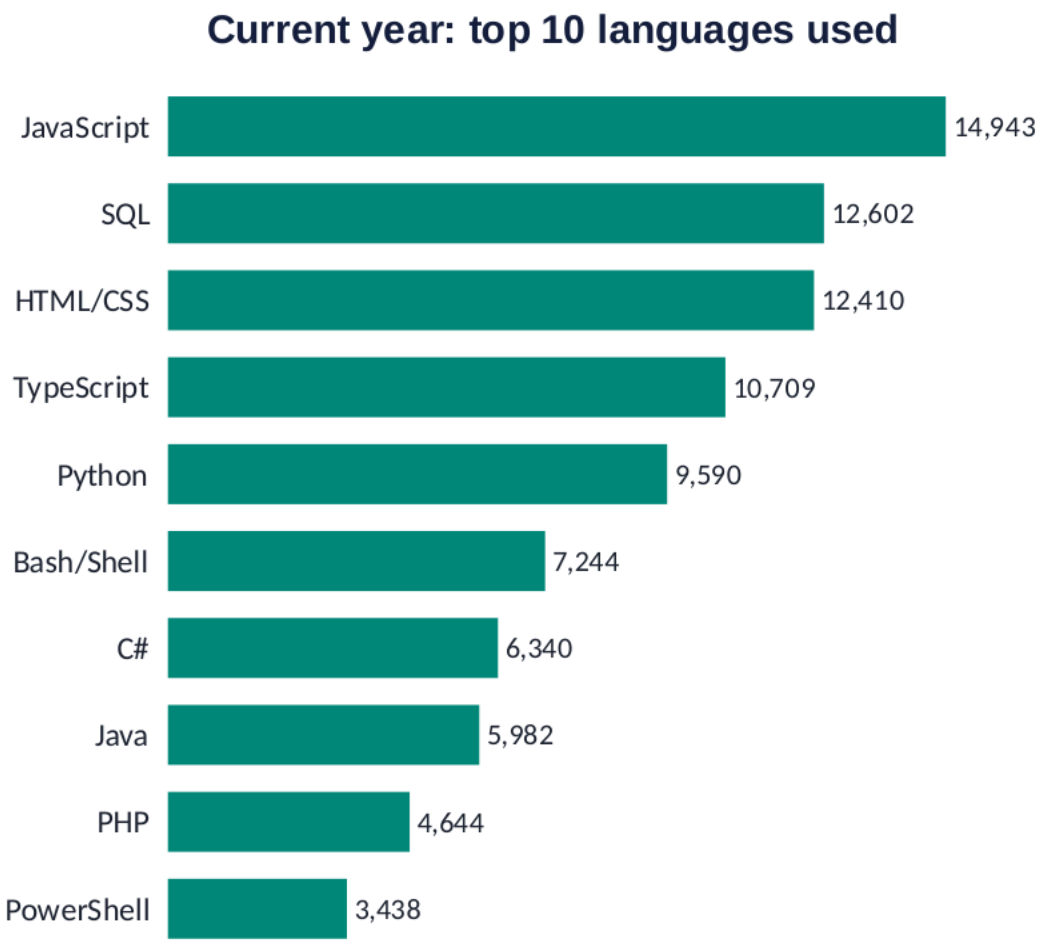
3 Analysis

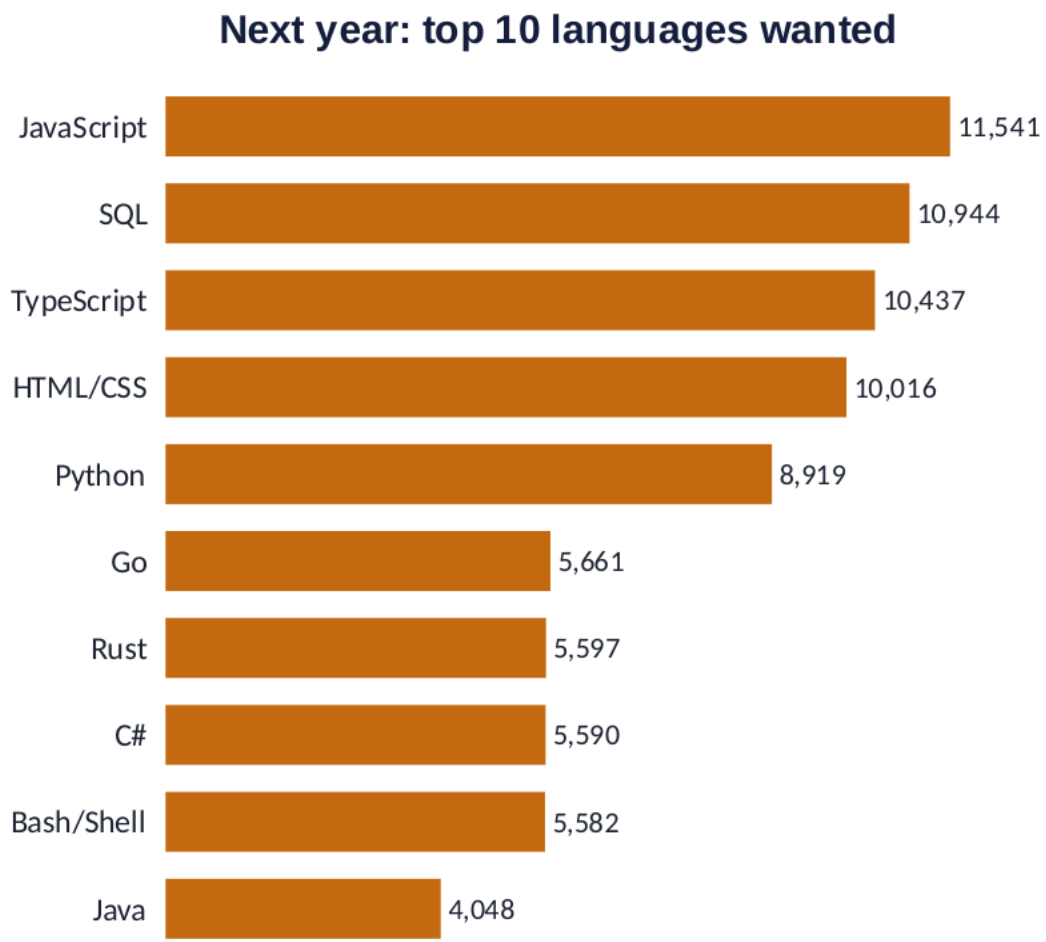
- Counted respondents per technology: used now vs. want next year
- Ranked top 10 and % share of respondents
- Momentum index = want ÷ use (techs with ≥1,000 users)

4 Visualization

- Python (pandas) for counts and charts
- IBM Cognos Analytics dashboard with 3 tabs
- Findings cross-checked between charts and dashboard

Programming Language Trends





Source: Stack Overflow Developer Survey 2024 subset provided by IBM Skills Network (n = 18,845). Multi-select answers split into single technologies.

Programming Language Trends: Findings & Implications

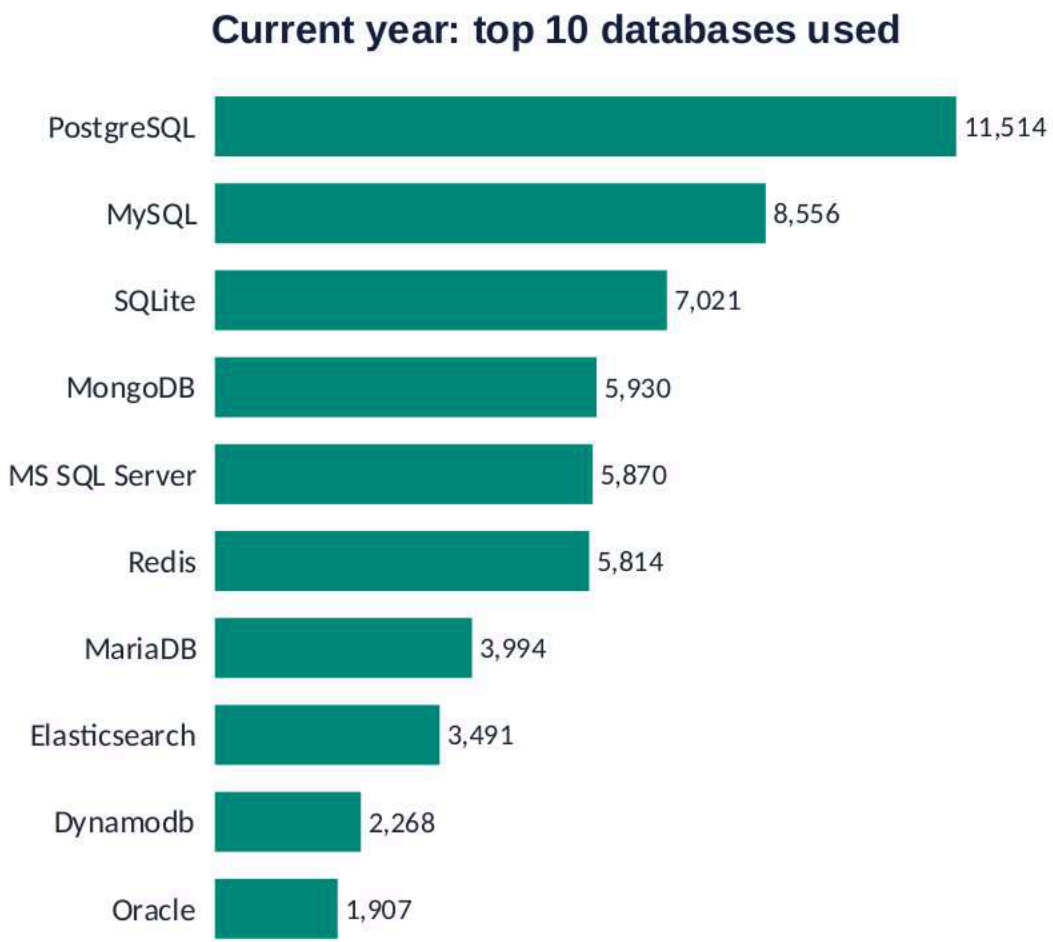
Findings

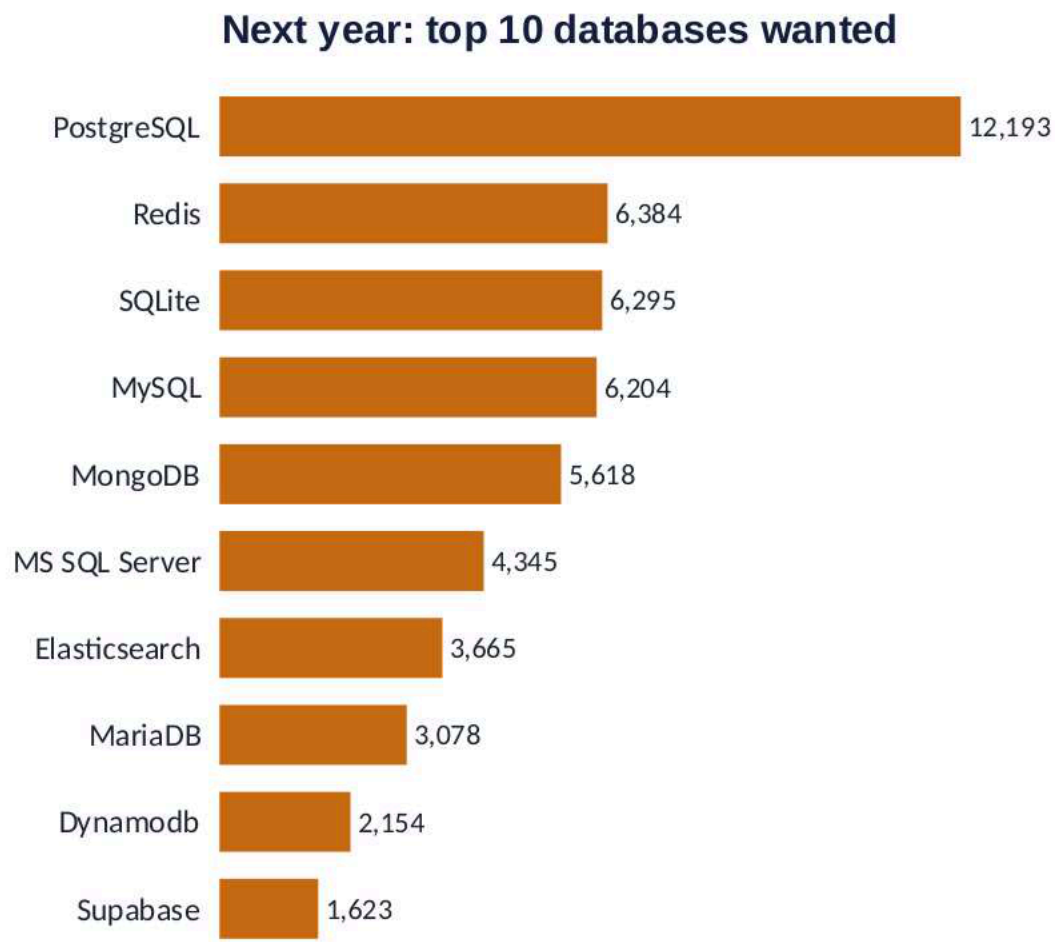
- JavaScript stays #1 both years, but its share falls from 79% (used) to 61% (wanted).
- TypeScript overtakes HTML/CSS in future demand (#3 vs. #4).
- Go (5,661) and Rust (5,597) enter the future top 10, replacing PHP and PowerShell.
- Highest momentum: Rust 2.45×, Go 1.71×, Kotlin 1.40×. Lowest: PHP 0.60×, Java 0.68×.

Implications

- JavaScript/TypeScript + SQL remain the baseline skill set for most developer roles.
- Go and Rust skills are worth building for cloud-native, systems and performance work.
- Python remains a safe top-5 choice across data, AI and back-end work.
- Teams built on PHP or Java may find hiring harder as interest shifts to newer languages.

Database Trends





Source: Stack Overflow Developer Survey 2024 subset provided by IBM Skills Network (n = 18,845). Multi-select answers split into single technologies.

Database Trends: Findings & Implications

61% → 65%

PostgreSQL share: used → wanted

#6 → #2

Redis rank: used → wanted

0.73×

MySQL momentum (falls #2 → #4)

1.45×

Supabase momentum (new in top 10)

Findings

- PostgreSQL is #1 today (11,514) and is wanted by even more developers next year (12,193).
- Redis rises to #2 in future demand (6,384); SQLite stays steady.
- MySQL (0.73×), MS SQL Server (0.74×) and Oracle (0.58×) all lose interest.
- Supabase replaces Oracle in the future top 10.

Implications

- PostgreSQL is the most transferable database skill to learn or hire for.
- Growth in Redis and Supabase points to caching, real-time and managed back-end services.
- Organisations on Oracle or MySQL should plan migration paths and retraining.
- Open-source databases are winning both current use and future intent.

Job Postings by Location



Key takeaways

- Washington DC leads with 5,316 postings (19.7% of 27,005).
- The top 5 cities hold 71% of all postings.
- San Francisco (435) and Austin (434) are low despite being tech hubs; Philadelphia has only 41.
- Job seekers should target DC, Detroit, Seattle, Houston and New York, or look at remote roles.

Source: job-postings.xlsx (13 US cities, 27,005 postings; "New Orleons" corrected to New Orleans).

Dashboard

An interactive dashboard was built in IBM Cognos Analytics from survey_data_updated.csv. It has three tabs, each using a 2 × 2 layout.



Current Technology Usage

Top 10 languages (bar), databases (column), platforms (word cloud) and web frameworks (bubble) respondents have worked with.

Slide 12



Future Technology Trends

Top 10 languages (bar), databases (column), platforms (tree map) and web frameworks (bubble) respondents want to work with.

Slide 13



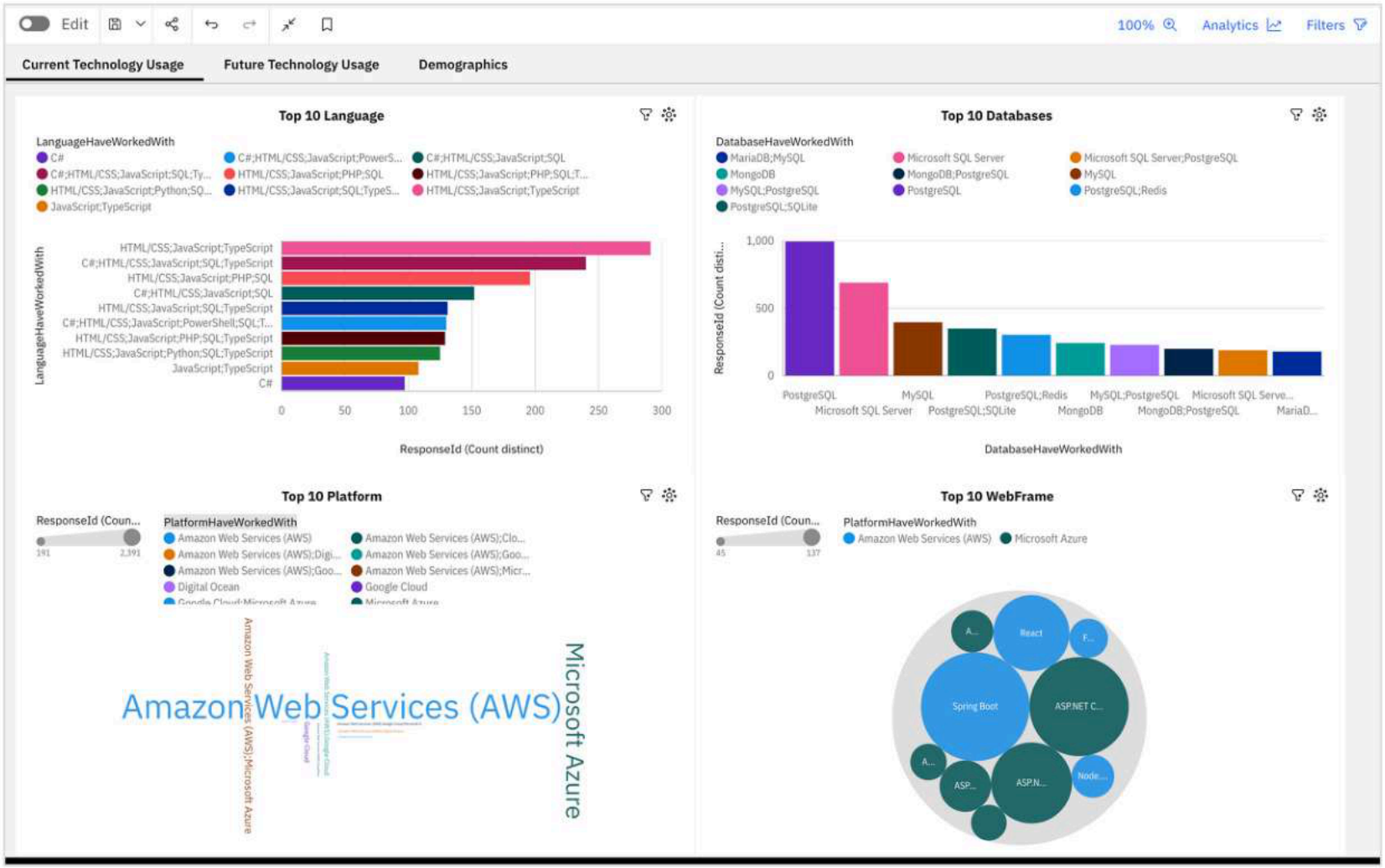
Demographics

Respondents by age (pie), country (map), education level (line) and age by education (stacked bar).

Slide 14

Note: the dashboard counts each respondent's full answer (e.g. "HTML/CSS;JavaScript;TypeScript"). Slides 6–9 count single technologies.

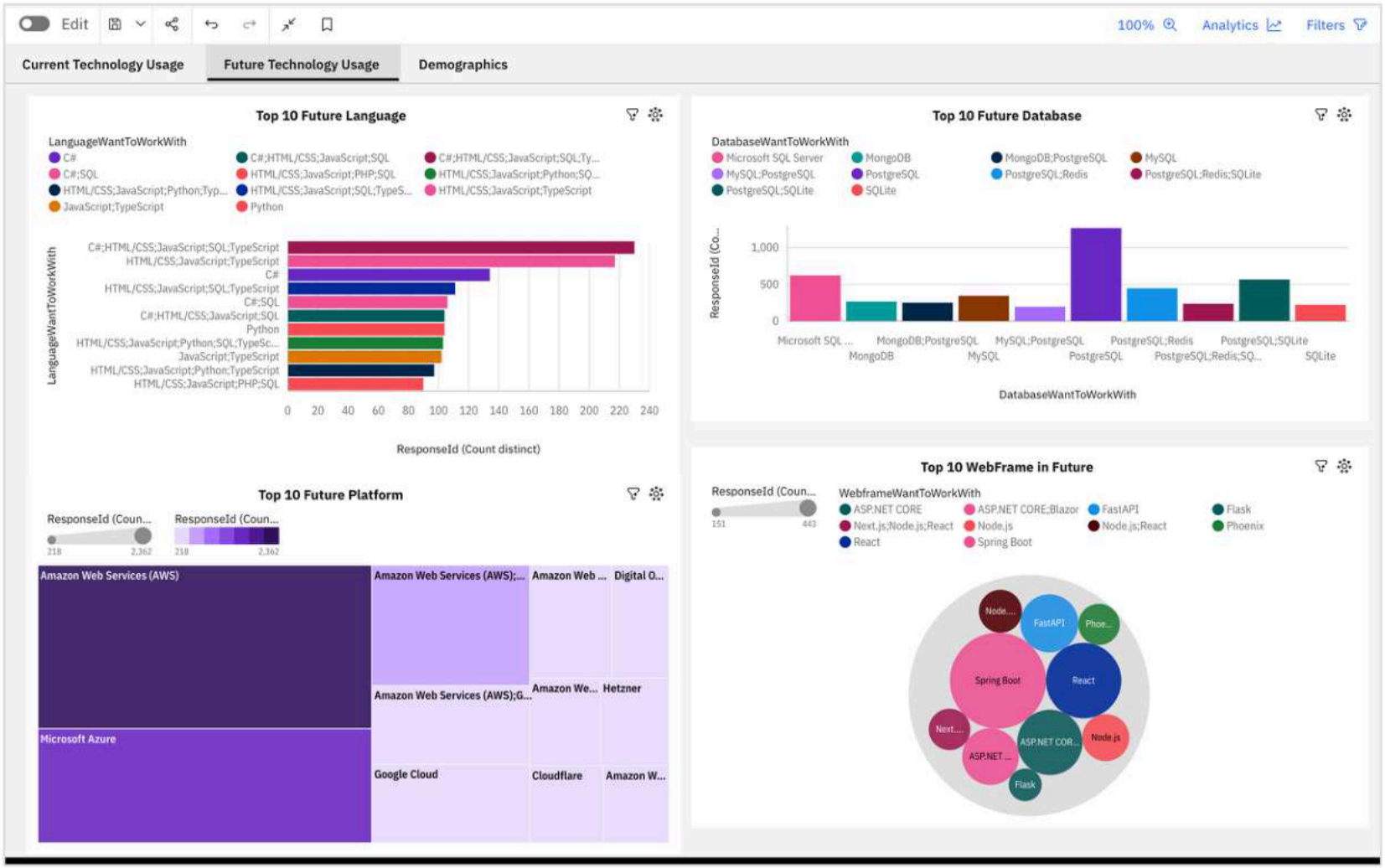
Dashboard: Current Technology Usage



What it shows

- Most common language stack: HTML/CSS + JavaScript + TypeScript (291)
- PostgreSQL alone is the top database answer (994)
- AWS and Azure dominate the platform word cloud

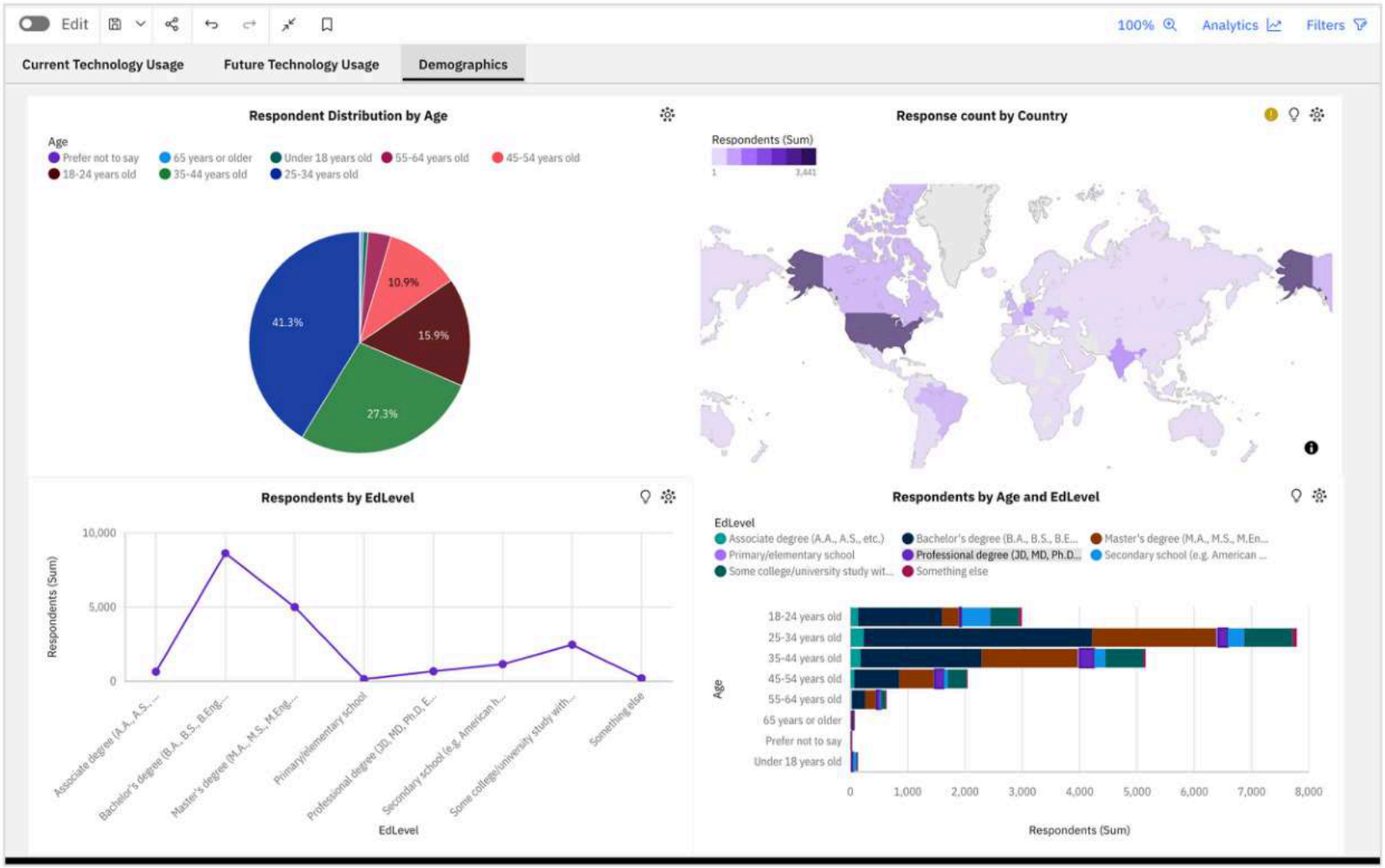
Dashboard: Future Technology Trends



What it shows

- C# + HTML/CSS + JS + SQL + TS becomes the most wanted stack (230)
- PostgreSQL alone rises to 1,265 wants
- AWS leads the tree map; Spring Boot and React lead frameworks

Dashboard: Demographics



What it shows

- 25–34 year-olds make up 41.3% of respondents
- The US is the largest country (3,441)
- Bachelor’s degree is the most common education level (8,629)

Dashboard Insights



Full-stack web is the norm

The top language answers are web stacks (HTML/CSS + JavaScript + TypeScript); C#/.NET full stack is #2 today and #1 next year.



PostgreSQL strengthens

As a single answer PostgreSQL grows from 994 to 1,265, matching its #1 rank in slides 8–9.



Multi-cloud is common

AWS alone (2,391) and Azure alone (1,780) lead; AWS + Azure together is the #3 answer (762).



Typical respondent

25–34 years old (41.3%) with a Bachelor's degree. This single segment has 3,985 people, the largest in the stacked bar.

Overall Findings & Implications

Findings

- Web foundations (JavaScript, SQL, HTML/CSS) are near-universal skills.
- Interest is moving to type-safe, performance-focused languages: TypeScript, Go and Rust.
- Open-source PostgreSQL is consolidating its lead; proprietary databases are losing interest.
- Cloud use concentrates on AWS, Azure and Google Cloud, while Cloudflare, Hetzner and Supabase grow.
- React and Node.js anchor the web; Svelte (1.73×) and FastAPI (1.14×) rise as jQuery (0.47×) fades.

Implications

Learners: build a JS/TS + SQL + PostgreSQL core, then add Python, Go or Rust.

Employers: expect more competition for Go, Rust and PostgreSQL talent; offer upskilling to keep staff on legacy stacks.

Industry context: matches wider trends. Rust has topped Stack Overflow's "most admired" list for years, PostgreSQL passed MySQL in 2023, and AWS leads cloud market share.

Conclusion

01 JavaScript, SQL and HTML/CSS dominate current developer work, and will still lead next year.

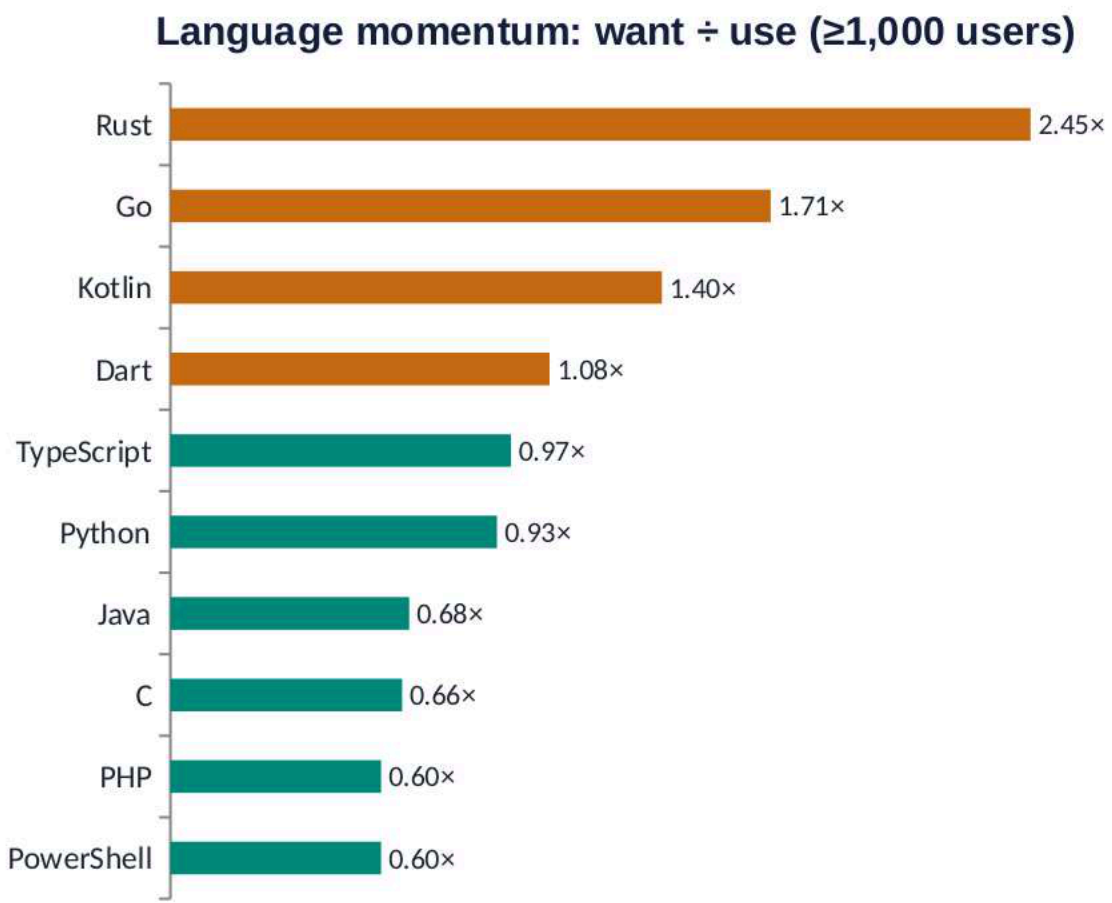
02 Go and Rust are the fastest-rising languages; PHP and Java are losing interest.

03 PostgreSQL is the database of choice now and next year; Redis and Supabase are rising.

04 AWS leads cloud; React and Node.js lead web frameworks.

05 Respondents are mostly 25–44, degree-educated, and based in the US, Germany and India.

Appendix: Momentum Index & Respondents by Country



Top 10 countries

Country	Respondents	Share
United States	3,441	18.3%
Germany	1,339	7.1%
India	1,316	7.0%
United Kingdom	1,053	5.6%
Ukraine	931	4.9%
Canada	661	3.5%
France	598	3.2%
Brazil	587	3.1%
Poland	494	2.6%
Netherlands	480	2.5%

Above 1.0× (amber): more developers want it than use it today. Below 1.0× (teal): interest is shrinking.

Source: Stack Overflow Developer Survey 2024 subset provided by IBM Skills Network (n = 18,845). Multi-select answers split into single technologies.